Replay-Aware TSP Benchmark Report

Overview

- Condition count: 9.
- Best final transfer gap: phase6_stratified_random_replay.
- Best TSPLIB holdout gap: phase6_stratified_random_replay.
- Best synthetic holdout gap: phase6_random_replay.

Run Metadata

- run_name: run_20260518_225158_b
- started_at_local: 2026-05-18 22:51:58
- finished_at_local: 2026-05-18 23:12:08
- duration_hhmm: 00:20
- duration_seconds: 1210.024
- seed_offset: 1000
- replicate_label: b
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_no_replay	none	score_only	False	0.120766	0.015789	0.082593	0.236772	0.142664	0.0	0.844353	0.76	0.007056
phase6_random_replay	random	score_only	False	0.184836	0.0	0.117623	0.236772	0.133657	0.19457	0.775482	0.76	-0.003091
phase6_failure_replay	failure	score_only	False	0.213387	0.064916	0.159398	0.171206	0.348017	0.261637	0.892563	0.56	-0.027903

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_random_replay_compression	random	novelty_gate	True	0.10109	0.000343	0.064455	0.236772	0.135611	0.19457	0.798751	0.76	0.01128
phase6_stratified_random_replay	stratified_random	score_only	False	0.096087	0.001628	0.061738	0.236772	0.183216	0.221162	0.673809	0.76	0.078344
phase6_diversity_weighted_replay	diversity_weighted	score_only	False	0.114454	0.0	0.072834	0.236772	0.193559	0.157984	0.588682	0.76	0.104505
phase6_residual_failure_replay	residual_failure	score_only	False	0.122954	0.0	0.078243	0.130778	0.291763	0.328165	0.712319	0.76	0.009337
phase6_diversity_failure_replay	diversity_failure	score_only	False	0.213387	0.064916	0.159398	0.171206	0.348017	0.17024	0.900373	0.56	-0.027661
phase6_diversity_failure_replay_compression	diversity_failure	novelty_gate	True	0.131164	0.010343	0.087229	0.171206	0.252962	0.173315	0.0	0.76	0.0

Condition Notes

phase6_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.120766, synthetic 0.015789, combined 0.082593.
- Accepted-epoch count 2, mean accepted code novelty 0.844353, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.142664, size bias 8.3e-05, and failure concentration 0.0.
- Replay selection diversity 0.0, mean selected expected gap 0.0, mean selected residual gap 0.0.
- Adaptation efficiency 0.007056 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 2, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.120766 across 7 instances; family means: ch=0.131699, kroD=0.065605, pcb=0.258557, pr=0.099844, rd=0.107585, st=0.05037.
- Panel synthetic_holdout mean gap 0.015789 across 4 instances; family means: clustered_gaussian=0.001371, grid_outliers=0.061786, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3406, "expected_gap": 0.40822, "family": "a", "name": "a280", "optimality_gap": 0.320667, "residual_gap": -0.087553}, {"best_known_cost": 629, "cost": 743, "expected_gap": 0.257552, "family": "eil", "name": "eil101", "optimality_gap": 0.18124, "residual_gap": -0.076312},

{"best_known_cost": 14379, "cost": 16371, "expected_gap": 0.322164, "family": "lin", "name": "lin105", "optimality_gap": 0.138535, "residual_gap": -0.183629}]

phase6_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.184836, synthetic 0.0, combined 0.117623.
- Accepted-epoch count 2, mean accepted code novelty 0.775482, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.133657, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.133575, mean selected residual gap -0.024041.
- Adaptation efficiency -0.003091 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 0, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.184836 across 7 instances; family means: ch=0.138289, kroD=0.286747, pcb=0.289397, pr=0.139619, rd=0.264475, st=0.037037.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3568, "expected_gap": 0.403955, "family": "a", "name": "a280", "optimality_gap": 0.383482, "residual_gap": -0.020473}, {"best_known_cost": 629, "cost": 774, "expected_gap": 0.25628, "family": "eil", "name": "eil101", "optimality_gap": 0.230525, "residual_gap": -0.025755}, {"best_known_cost": 14379, "cost": 16927, "expected_gap": 0.329244, "family": "lin", "name": "lin105", "optimality_gap": 0.177203, "residual_gap": -0.152041}]

phase6_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.892563, and final complexity 0.56.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.348017, size bias 0.120322, and failure concentration 0.261637.
- Replay selection diversity 0.177669, mean selected expected gap 0.308457, mean selected residual gap 0.0673.
- Adaptation efficiency -0.027903 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.403722, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.002404}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.25469, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.136407}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.328841, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": -2.8e-05}]

phase6_random_replay_compression

- Replay mode: random with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.10109, synthetic 0.000343, combined 0.064455.

- Accepted-epoch count 2, mean accepted code novelty 0.798751, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.135611, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.132418, mean selected residual gap -0.022565.
- Adaptation efficiency 0.01128 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 1, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.10109 across 7 instances; family means: ch=0.081514, kroD=0.122523, pcb=0.259876, pr=0.053782, rd=0.053603, st=0.054815.
- Panel synthetic_holdout mean gap 0.000343 across 4 instances; family means: clustered_gaussian=0.001371, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3357, "expected_gap": 0.402947, "family": "a", "name": "a280", "optimality_gap": 0.301667, "residual_gap": -0.10128}, {"best_known_cost": 14379, "cost": 17085, "expected_gap": 0.328006, "family": "lin", "name": "lin105", "optimality_gap": 0.188191, "residual_gap": -0.139815}, {"best_known_cost": 629, "cost": 737, "expected_gap": 0.253418, "family": "eil", "name": "eil101", "optimality_gap": 0.171701, "residual_gap": -0.081717}]

phase6_stratified_random_replay

- Replay mode: stratified_random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.096087, synthetic 0.001628, combined 0.061738.
- Accepted-epoch count 3, mean accepted code novelty 0.673809, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.183216, size bias 8.3e-05, and failure concentration 0.221162.
- Replay selection diversity 0.187367, mean selected expected gap 0.237428, mean selected residual gap -0.027686.
- Adaptation efficiency 0.078344 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.096087 across 7 instances; family means: ch=0.061645, kroD=0.101813, pcb=0.228209, pr=0.061169, rd=0.078129, st=0.08.
- Panel synthetic_holdout mean gap 0.001628 across 4 instances; family means: clustered_gaussian=0.006511, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3436, "expected_gap": 0.402171, "family": "a", "name": "a280", "optimality_gap": 0.332299, "residual_gap": -0.069872}, {"best_known_cost": 7542, "cost": 9649, "expected_gap": 0.233068, "family": "berlin", "name": "berlin52", "optimality_gap": 0.279369, "residual_gap": 0.046301}, {"best_known_cost": 14379, "cost": 17018, "expected_gap": 0.326657, "family": "lin", "name": "lin105", "optimality_gap": 0.183532, "residual_gap": -0.143125}]

phase6_diversity_weighted_replay

- Replay mode: diversity_weighted with selection mode score_only.
- Final transfer gaps: TSPLIB 0.114454, synthetic 0.0, combined 0.072834.
- Accepted-epoch count 2, mean accepted code novelty 0.588682, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.193559, size bias 8.3e-05, and failure concentration 0.157984.
- Replay selection diversity 0.267365, mean selected expected gap 0.154023, mean selected residual gap 0.008469.

- Adaptation efficiency 0.104505 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.114454 across 7 instances; family means: ch=0.138782, kroD=0.110172, pcb=0.226318, pr=0.070988, rd=0.098357, st=0.017778.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3277, "expected_gap": 0.402947, "family": "a", "name": "a280", "optimality_gap": 0.270648, "residual_gap": -0.132299}, {"best_known_cost": 7542, "cost": 9544, "expected_gap": 0.235826, "family": "berlin", "name": "berlin52", "optimality_gap": 0.265447, "residual_gap": 0.029621}, {"best_known_cost": 14379, "cost": 17487, "expected_gap": 0.326657, "family": "lin", "name": "lin105", "optimality_gap": 0.216149, "residual_gap": -0.110508}]

phase6_residual_failure_replay

- Replay mode: residual_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.122954, synthetic 0.0, combined 0.078243.
- Accepted-epoch count 2, mean accepted code novelty 0.712319, and final complexity 0.76.
- Active replay archive residual_archive with mean diversity 0.130778, mean hardness 0.291763, size bias 0.080581, and failure concentration 0.328165.
- Replay selection diversity 0.140953, mean selected expected gap 0.289892, mean selected residual gap 0.046059.
- Adaptation efficiency 0.009337 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.122954 across 7 instances; family means: ch=0.108439, kroD=0.28717, pcb=0.224349, pr=0.015653, rd=0.082554, st=0.034074.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3271, "expected_gap": 0.405273, "family": "a", "name": "a280", "optimality_gap": 0.268321, "residual_gap": -0.136952}, {"best_known_cost": 14379, "cost": 16015, "expected_gap": 0.324988, "family": "lin", "name": "lin105", "optimality_gap": 0.113777, "residual_gap": -0.211211}, {"best_known_cost": 629, "cost": 674, "expected_gap": 0.262003, "family": "eil", "name": "eil101", "optimality_gap": 0.071542, "residual_gap": -0.190461}]

phase6_diversity_failure_replay

- Replay mode: diversity_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.900373, and final complexity 0.56.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.348017, size bias 0.120322, and failure concentration 0.17024.
- Replay selection diversity 0.199284, mean selected expected gap 0.301218, mean selected residual gap 0.04877.
- Adaptation efficiency -0.027661 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.

- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.405351, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.004033}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.267091, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.124006}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.325433, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.00338}]

phase6_diversity_failure_replay_compression

- Replay mode: diversity_failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.131164, synthetic 0.010343, combined 0.087229.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.252962, size bias 0.120322, and failure concentration 0.173315.
- Replay selection diversity 0.199207, mean selected expected gap 0.307214, mean selected residual gap -0.05521.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.131164 across 7 instances; family means: ch=0.10119, kroD=0.181366, pcb=0.218815, pr=0.131972, rd=0.096207, st=0.087407.
- Panel synthetic_holdout mean gap 0.010343 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.041372, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3258, "expected_gap": 0.405739, "family": "a", "name": "a280", "optimality_gap": 0.26328, "residual_gap": -0.142459}, {"best_known_cost": 629, "cost": 748, "expected_gap": 0.263911, "family": "eil", "name": "eil101", "optimality_gap": 0.189189, "residual_gap": -0.074722}, {"best_known_cost": 7542, "cost": 8312, "expected_gap": 0.244206, "family": "berlin", "name": "berlin52", "optimality_gap": 0.102095, "residual_gap": -0.142111}]

Judge Appendix

Summary of Key Results

Condition	Replay Type	Transfer (Held-out TSPLIB Gap)	Synthetic Transfer Gap	Synthetic Holdout Gap	Final Training Gap (Last)	Adaptation Efficiency	Replay Failure Concentration	Archive Diversity	Archive Hardness	Code Novelty (mean)	Notes
phase6_no_replay	None	0.1208	0.0826	0.0158	0.1751	0.0071	0.0	0.237	0.143	0.844	Baseline, moderate gaps
phase6_random_replay	Raw-failure random	0.1848	0.1176	0.0	0.2364	-0.0031	0.1946	0.237	0.134	0.775	Higher gaps, some failure concentration
phase6_failure_replay	Raw-failure from worst archive	0.2134	0.1594	0.0649	0.3602	-0.0279	0.262	0.172	0.348	0.893	Highest gaps and hardness, high failure concentration
phase6_random_replay_compression	Raw-failure with compression	0.1011	0.0645	0.0003	0.1866	0.0113	0.195	0.237	0.136	0.799	Improved transfer gaps and synthetic gap with compression+random replay

Condition	Replay Type	Transfer (Held-out TSPLIB Gap)	Synthetic Transfer Gap	Synthetic Holdout Gap	Final Training Gap (Last)	Adaptation Efficiency	Replay Failure Concentration	Archive Diversity	Archive Hardness	Code Novelty (mean)	Notes
phase6_stratified_random_replay	Stratified random replay	**0.0961**	**0.0617**	0.0016	0.2393	0.0783	0.221	0.237	0.183	0.674	Best transfer performance; moderate hardness
phase6_diversity_weighted_replay	Diversity-weighted replay	0.1145	0.0728	0.0	0.3012	0.1045	0.158	0.240	0.198	0.589	High adaptation efficiency with diversity-weighted replay
phase6_residual_failure_replay	Residual failure replay	0.1230	0.0782	0.0	0.3602	0.0093	0.262	0.201	0.372	0.712	Moderate transfer gaps, higher hardness and failure concentration
phase6_diversity_failure_replay	Diversity failure replay	0.2134	0.1594	0.0649	0.3602	-0.0277	0.170	0.175	0.349	0.900	Similar to pure failure replay, high gap and failure concentration
phase6_diversity_failure_replay_compression	Diversity failure replay + compression	0.1312	0.0872	0.0103	0.1599	0.0	0.173	0.175	0.307	0.0	Stability in code novelty, gaps intermediate

Interpretation

- **Best transfer performance** (lowest gaps on held-out TSPLIB and synthetic holdout instances) is observed for **phase6_stratified_random_replay** (held-out TSPLIB gap 9.61%, synthetic 0.16%) with good adaptation efficiency (0.0783) and moderate failure concentration (0.22). This indicates broad-coverage replay with stratification enhances generalization to unseen instances.

- **Raw-failure and diversity failure replay modes** (phase6_failure_replay, phase6_diversity_failure_replay) increase archive hardness and failure concentration but suffer higher gaps on transfer benchmarks (~21-22% on TSPLIB). Their adaptation efficiency is negative, showing diminished learning efficiency. These conditions show failure concentration around 0.26-0.27, consistent with replaying mostly hard cases but poor generalization.

- **Compression combined with random or diversity-weighted replay** improves transfer gaps noticeably compared to uncompressed failure replay (e.g., phase6_random_replay_compression has 10.1% TSPLIB gap vs. 18.5% in phase6_random_replay). This suggests compression facilitates focusing on useful replay cases without increasing failure concentration or harming diversity.

- **Code novelty** (mean and last) tends to be **lower in best transfer cases**: phase6_stratified_random_replay has code novelty ~0.67 vs. ~0.9 in failure-based replay. This decouples lexical novelty from algorithmic improvement and shows code novelty alone is not correlated with transfer performance here.

- **Replay archive diversity** is highest (~0.24) except in failure replay modes where it drops (~0.17-0.20), indicating replay diversity positively correlates with transfer quality.

- **Size bias** in archives is negligible for most but higher in failure replay (~0.12-0.13), consistent with bias toward harder cases.

- **Final synthetic holdout gaps** are minimal (<0.002) in stratified and random replay conditions (including compressed versions), showing effective generalization in synthetic domain.

Mechanism Insights

- **Diversified replays** (stratified random, diversity-weighted) yield broad coverage, better transfer gaps, balanced hardness (~0.18–0.20), and lower failure concentration (~0.16–0.22).

- **Failure replay modes** focus on hardest or residual-failure cases, increasing hardness and failure concentration but resulting in overfitting and poor transfer.

- **Compression pressure** when combined with replay (random or diversity) reduces synthetic and transfer gaps without increasing failure concentration, indicating efficient archive management.

Summary

- The **phase6_stratified_random_replay** condition offers the best balance of transfer generalization (lowest gaps), moderate archive hardness, and archive diversity, dominating other replay modes.

- Failure-based replay conditions increase replay failure concentration and hardness but degrade transfer performance substantially; indicative of overfitting on difficult cases.

- Compression techniques enhance replay effectiveness in random or diversity-weighted replays by maintaining archive quality without loss of diversity.

- Code novelty decreases in better transfer conditions relative to higher-novelty failure replay modes, highlighting that lexical novelty is not a substitute for algorithmic innovation.

In conclusion, broad-coverage replay with stratified random sampling and compression yields superior transfer performance over failure-focused or diversity failure replay strategies, despite the latter showing higher code novelty and replay hardness.